ITE4116M FINAL YEAR PROJECT

PROJECT SUGGESTION FORM

Academic Year 2026/27

Submission Guide: Students must form a group of 3 to 4 members, discuss the project topic with your preferred supervisor, and obtain verbal agreement. Once agreed, the group must complete this Project Suggestion Document and upload the document file (**in both PDF & DOCX**) on Moodle during the FYP session by **14-SEP-2026.**

1. General Project Information

Proposed Project Title	AI and IoT Based Food and Nutrition Monitoring System
Assigned Project Supervisor	Marco (revised scope for review)
Programme / Class Code	Higher Diploma in AI and Smart Technology (IT114124) / Class AIST-2B
Student Group Number	Group 2

Student Member Details (Normally 3 to 4 members):

S/N	Student Name	Student ID	Email Address	Proposed Major Role / Area of Responsibility
1	Yuan Yuchen	250342840	250342840@stu.vtc.edu.hk	**Project Leader / Backend AI Lead (40%)** Requirements, APIs, food recognition, nutrition estimates, forecasts and integration tests.
2	Xu Ziqian	250137555	250137555@stu.vtc.edu.hk	**IoT Hardware / Edge Processing (30%)** Sensing hardware, Pi acquisition, container calibration, event detection and reliability tests.
3	Li Wenhua	250329909	250329909@stu.vtc.edu.hk	**Web Application / UX Developer (30%)** Scenario dashboards, QR menu/orders, nutrition views, bilingual UI and usability tests.

2. Problem Statement & Project Objectives

Instructions: Describe the business or technical problem in information technology that this project intends to address. Define 2-3 specific project objectives (what your solution intends to achieve for the target users/businesses).

Problem Statement:
Foodservice teams, schools and individuals need reliable links between food quantity, nutritional information and leftovers. Buffets need timely replenishment, schools need visibility of meal portions and food-group balance, and individuals need easier meal records.
We propose a configurable system that continuously monitors food while serving containers or meal plates remain on sensing bases. Measured quantities, food recognition and reference nutrition data support replenishment, school meal review and personal nutrition awareness. Metafoodx and Controliza inform the design; the prototype will evaluate these uses through controlled scenarios.

Project Objectives:
1. Build a reusable IoT and vision prototype for continuous monitoring of food identity, quantity and leftovers in serving and individual-meal configurations.
2. Provide AI-assisted replenishment and reference-based portion/nutrition feedback through foodservice, school and personal profiles, retaining human confirmation and uncertainty indicators.
3. Evaluate measurement, recognition, nutrition calculations, replenishment outcomes and usability against manual references and simple baselines in controlled scenarios.

3. Proposed Project Scope & Key Features

Instructions: Define the scope of your system. List and briefly describe the key modules, functions, or features that your group plans to implement. Projects must be implementation-oriented and student-centered.

Key Subsystems / Features:
A. Shared continuous sensing. Track weight, dish identity, refills and leftovers; handle tare, movement, occlusion and stale data. Record surface temperature where applicable.
B. Nutrition estimates. Link measured portions to registered recipes or matched food-composition records. Separate food provided, leftovers and user-confirmed intake estimates; flag uncertain food/weight mappings.
C. Three application profiles. Foodservice: replenishment timing and batch tasks. Schools: meal-portion/food-group review and aggregate leftovers. Personal: adult meal records and reference nutrition summaries.
D. Interfaces and feedback. Bilingual dashboards, alerts and reports. Retain QR menus with live availability, completed-order popularity and estimated waits; basic ordering and kitchen status remain available in foodservice profiles.
Prototype Scope: One shared six-month prototype, up to three weighing positions and 4-6 registered foods/recipes. Demonstrate three profiles using separately weighed items or known recipes and anonymous school example meals.

Expected Users & Roles: Kitchen staff, school catering teams/educators and adult personal users; parents/students can view appropriate menu and educational summaries.

4. Technical Stack & Special Requirements

Instructions: Detail the hardware and software tools you intend to use. Highlight how your project incorporates contemporary technologies (AI, IoT, Cloud, etc.) and how it aligns with modern considerations like environmental-friendly design, bilingual/multi-language support, or cross-border deployability to meet Greater China business needs.

Development Technologies & Tools:

- **Hardware:** Raspberry Pi 5, load cells with ADC interfaces, overhead camera, surface-temperature sensor and display/LEDs. Adapt supports for serving containers or meal plates; assess RGB-D by testing. Existing UNO R4 WiFi is optional.
- **Programming Languages:** Python, TypeScript, HTML/CSS; C++ only if a microcontroller is used.
- **Frameworks / Libraries:** OpenCV, PyTorch/scikit-learn, FastAPI, React and i18next.
- **Database / Storage:** PostgreSQL, local logs, selected USDA FoodData Central records and registered recipes. Retain source IDs, units and cooking-state/recipe metadata.
- **Target Platforms:** Local Linux device and responsive web interfaces with configurable scenario profiles.

Special Project Considerations:

- **Contemporary Technologies:** IoT sensing, AI food recognition and demand estimates, reference-based nutrition calculations and live feedback; optional cloud backup.
- **Environmental-Friendly Solution / Social Benefit:** Support school meal review, personal nutrition awareness and avoidable-waste reduction with reusable hardware.
- **Bilingual / Multi-Language Support:** English and Traditional Chinese interfaces.
- **Cross-Border Deployment / Greater China Needs:** Configurable food catalogues, meal plans, currency and time zone; reviewed regional guidance, such as Hong Kong school-lunch guidelines.
- **Responsible Use:** Educational estimates, not diagnosis or prescribed diets. Uncertain mixed meals require confirmation; school demonstrations use anonymous meals without child health profiles.
- **Feasibility / Validation:** Test sensing and nutrition calculations against known portions/manual references; compare replenishment with fixed-threshold and consumption-rate baselines. Use controlled meals and adult usability tests; field access is not assumed. Full payment/ERP integration is excluded.

5. Endorsements & Supervisor Feedback

Student Group Endorsement

By submitting this suggestion, we confirm that all listed group members are in agreement regarding this project concept and our assigned roles.

Student Representative Signature: _____

Date: _____

Supervisor Vetting Decision

Supervisor, please put a ☑ in the appropriate box.

☐ **APPROVED — The group is authorized to proceed to draft their formal Project Proposal.**

☐ **REJECTED — The group must modify their project topic suggestion and resubmit for review.**

Supervisor Feedback / Comments

Supervisor Endorsement

Project Supervisor Signature: _____

Date: _____